

Building the Enabling Environment for a Market for Quality

A policy brief for government, regulators, and the Uganda Dairy Development Authority

Based on "When Quality (doesn't) Pay: Evidence from Two Experiments in Ugandan Dairy" (Van Campenhout, Ariong, Kariuki, and Chamberlin). July 2026.

The policy problem

Uganda's dairy sector is a growth success on paper. Formal dairy exports rose from about USD 131.5 million in 2018 to USD 264.5 million in 2022/23. Yet the domestic value chain that feeds those exports still does not reward quality. Farmers who deliver cleaner, richer milk are paid the same per liter as those who do not, and adulteration remains common.

Our two field experiments identify why, and the answer is uncomfortable for the standard policy response of farmer training. The binding constraint is not that farmers cannot produce quality. It is that the market gives no one a reason to pay for it, and where a price does appear, the structure of intermediation decides who keeps it. Technology and training alone will not close this gap. Government has to shape the market architecture around three things: make quality observable to everyone, create downstream demand that pays for it, and ensure enough competition that the reward reaches farmers.

What the evidence shows

Observability is necessary but not sufficient. Placing milk analyzers and digital records at collection centers (a randomized trial across 95 catchment areas) sharply reduced adulteration: deliveries with more than 5 percent added water fell from 11 percent to 5 percent. But no price premium for quality emerged anywhere in the chain, because processors buy at uniform per-liter rates and transmit no demand for quality upstream. Measurement disciplined the worst behavior; it did not create a market.

Prices do not automatically trickle down. In a second experiment we introduced the missing price directly, paying quality bonuses to traders. Quality did improve once the standard was demanding enough to bind. But the surplus stayed with the traders. Farmers who supplied bonus-receiving traders actually received about 17 UGX per liter less, not more, in part because the bonus was paid privately and farmers never knew it existed, so they had no standing to demand a share. A price signal injected into a concentrated intermediation layer can be captured, and can even leave farmers worse off.

Together these results define the policy agenda. Fixing measurement is the easy part, and much of it can be left to firms. The public role is to build the surrounding institutions that turn observable quality into a reward that actually reaches producers.

Seven recommendations

- 1. Publish transparent standards and a reference quality-pricing schedule.** Issue and regularly update clear minimum compositional standards and a recommended premium grid for butterfat and solids-not-fat (SNF). A published focal point coordinates the market and nudges buyers toward quality-based pricing without heavy-handed mandates. This is a low-cost, high-leverage first move.
- 2. Establish neutral public testing points.** Fund or license independent, third-party milk-testing stations, the quality analog of a public weighbridge, where any farmer or trader can have milk tested cheaply and credibly. This breaks the current situation in which the buyer both sets the price and controls the only measurement, and it levels the playing field for quality discovery across large and small players alike.
- 3. Create a binding dispute and arbitration mechanism.** When a farmer and a buyer disagree over quality, an independent referee test at a public testing point should settle it, with penalties for buyers who are found to systematically under-grade. In our data, opaque measurement let intermediaries discount milk they claimed was substandard and pocket the difference. A credible right of appeal removes that rent and protects the honest producer.
- 4. Increase competition among traders and buyers.** Pass-through happens only when farmers have somewhere else to sell. Lower barriers to entry for collection agents, support farmer cooperatives and aggregation platforms, and enable digital marketplaces that widen farmers' outside options. The more buyers compete for good milk, the more of any quality premium ends up in farmers' hands.
- 5. Guarantee farmers a right to their own quality reading.** Require that any test result or quality-based bonus be disclosed to the producer, for example through a mandatory short message service (SMS) receipt at the point of sale. The negative pass-

through we observed was possible only because farmers were kept in the dark. Transparency here is a direct, pro-poor lever that costs the state almost nothing.

6. Use public demand as an anchor for quality. Government-linked buyers, school feeding, hospitals, the military, and export certification schemes, should purchase on graded quality and pay differentiated prices. This manufactures the downstream demand for quality that the market currently lacks, and pulls quality premiums up the chain rather than waiting for private processors to move first.

7. Coordinate seasonal, dynamic quality floors. Work with processors to tighten quality standards in the flush season, when milk is abundant and adulteration incentives peak, and to relax them in the dry season. A coordinated seasonal floor stabilizes both quality and farm-gate prices and prevents gluts of watered milk from depressing the whole market.

Bonus recommendation. Tie public support to verified quality. Link subsidized inputs, credit, and extension to verified quality delivery. This aligns the farmer investment that programs already fund with the market standard, so public spending on production reinforces, rather than runs ahead of, the emerging price signal.

Bottom line

The sequence matters. Observability comes first, but on its own it stalls at the intermediation layer, exactly as our experiments show. Government's distinctive contribution is to supply what firms will not: neutral measurement and dispute resolution, competition that forces pass-through, transparency that protects farmers, and public demand that pays for grade. Get that architecture right and quality upgrading becomes self-sustaining. Leave it to technology and training alone and the returns will keep accruing to the middle of the chain, not to the farmers the policy is meant to serve.